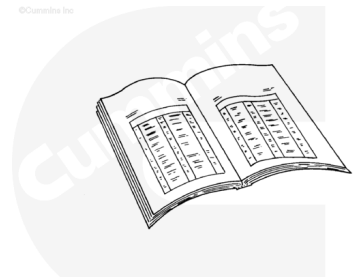


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

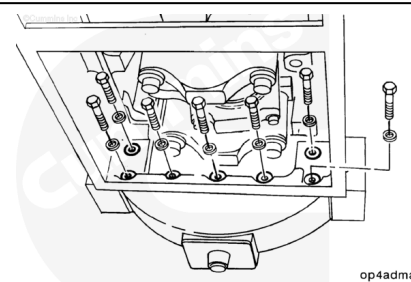
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/20/20-016-005-tr.html>)

- Drain the oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)
- Remove the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-024-tr.html>)
- Remove the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/20/20-013-020-tr.html>)
- Remove the adapter cover plate or oil pan (whichever is in the front position). Refer to Procedure 007-026 in Section 7 (</qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html>) or Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-025-tr.html>)
- Remove the rear engine mounts from the flywheel housing. Refer to Procedure 016-010 in Section 16. (</qs3/pubsys2/xml/en/procedures/20/20-016-010.html>)

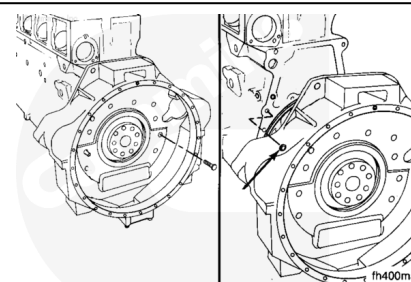
Remove

Remove the two [7/16-14 inch] capscrews, and the five [3/8-16 inch] capscrews from the flywheel housing.



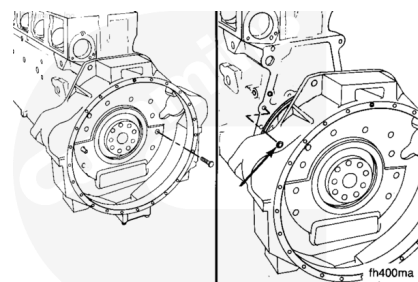
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

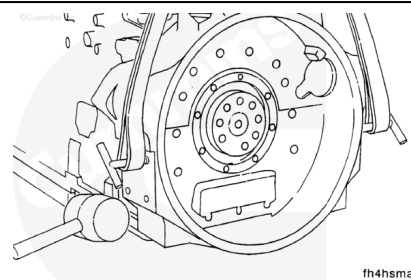


Use two guide studs to prevent the flywheel housing from rotating during disassembly. Remove two capscrews. Install the guide studs.

Use a hoist, a tee handle, and a lifting sling. Install the tee handle. Adjust the hoist until there is tension in the lifting sling. Remove the remaining capscrews.



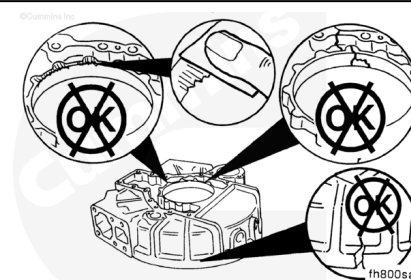
Use a mallet and tap the flywheel housing off the two locating dowels. Remove and discard the rectangular seal and the bolt seals.



Inspect for Reuse

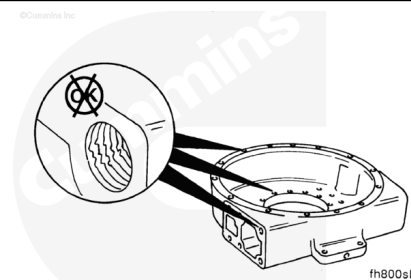
Inspect all surfaces for nicks, burrs, or cracks.

Use a fine crocus cloth to remove small nicks and burrs.



Inspect all threaded capscrew holes for damage.

Repair or replace the housing if the capscrew holes are damaged.

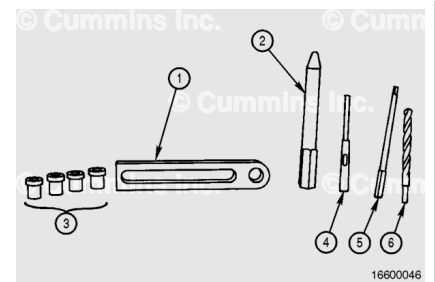


Redowel

The tools needed to perform this procedure are:

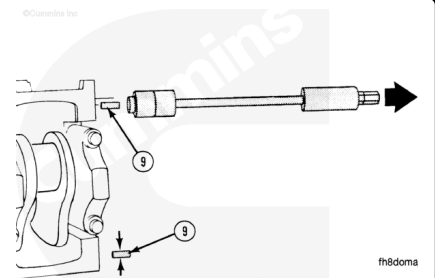
Drill Ream Fixture, Part Number ST-1232, that contains:

1. Plate, Part Number ST-1232-1, or equivalent
2. Locator Pin, Part Number 3375052, or equivalent
3. Drill/Ream Bushing Set (actual sizes depend on the dowel size as listed in this Procedure.)
4. Drill Adapter (locally obtained; use to adapt open (1) shank reamers to drill-chuck)
5. Reamer (locally obtained)
6. Drill Bit (locally obtained).



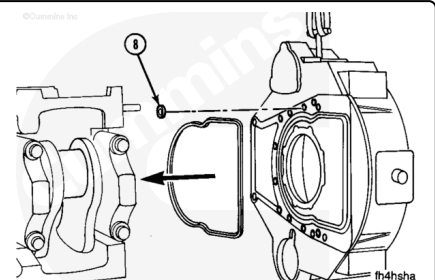
Use a dowel pin extractor, Part Number 3163720, or equivalent. Remove the two dowels (9) from the block.

Measure a dowel pin that is removed so that an oversize dowel pin can be determined.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



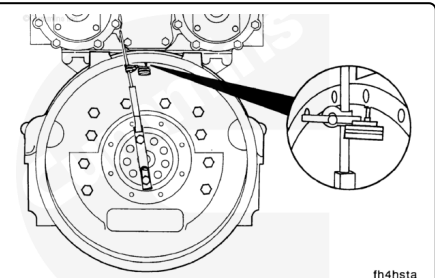
Note : Do not torque the capscrews. The flywheel housing must be aligned first.

Install the new rectangular seals (8) in the flywheel housing.

Align the flywheel housing to the crankshaft.

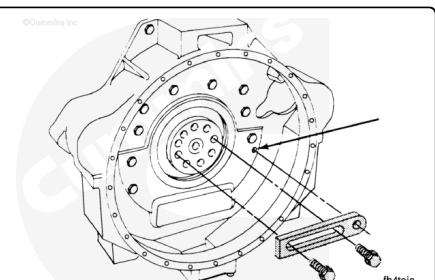
Move the housing with a mallet until the bore is within specification. Check to make sure that the face of the housing is in alignment. When the housing is in alignment, tighten the capscrews.

After the capscrews are tightened, check the alignment again.



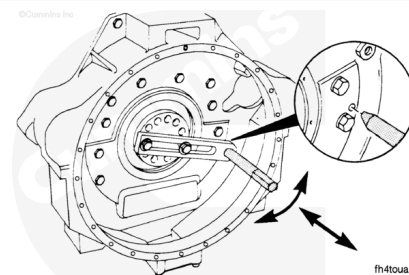
Use the appropriate size capscrews. Attach the plate, Part Number ST-1232-1, or equivalent, that is contained in drill ream fixture, Part Number ST-1232, or equivalent.

Hand-tighten the capscrews so the plate can be moved.



⚠ CAUTION ⚠

Make sure the crankshaft is in the LOCKED position during reaming to prevent damage to the engine.



Use the locator pin to align the plate with the hole for the dowel pin.

Tighten the capscrews. The taper on the pin **must** engage the dowel pin hole.

The locator pin **must** rotate easily after the capscrews are tightened.

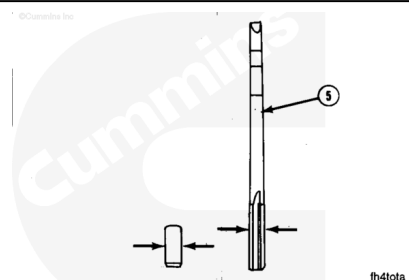
Lock the crankshaft in position. Check to make sure the locator pin is still in alignment and that the locator pin can be rotated easily.

Measure the dowel pins to be installed.

Obtain a reamer (5) that is 0.13 mm to 0.02 mm [0.005 in to 0.001 in] smaller than the dowel.

The dowel **must** be long enough to protrude from the block one-half of the flywheel housing wall thickness.

Note : There are three oversize dowel pins available from Cummins Inc.

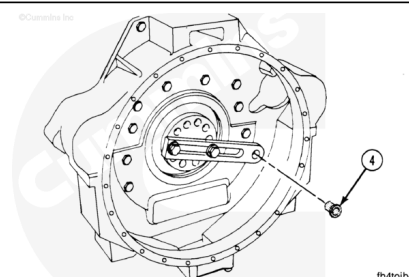


Oversize Dowel Pin Outside Diameter	
Overall Dimension	Oversize
13.08 mm [0.515 in]	0.38 mm [0.015 in]
13.46 mm [0.530 in]	0.76 mm [0.030 in]
13.84 mm [0.545 in]	1.14 mm [0.045 in]

Install the appropriate drill bushings (4). The table shows the bushings available from Cummins Inc.

Drill/Ream Bushing Sets – 25.4 mm [1 Inch] Outside Diameter

Oversize		Bushing Size	
mm	[in]	mm	[in]
3376495	Special	12.304	[0.4844]
	Standard	12.700	[0.5000]
0.38	[0.015]	13.096	[0.5156]



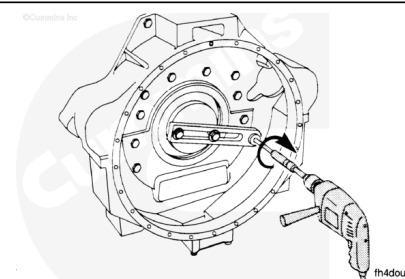
Drill/Ream Bushing Sets – 25.4 mm [1 Inch] Outside Diameter

0.76	[0.30]	13.494	[0.5312]
1.14	[0.045]	13.879	[0.5464]
ST-1234	Standard	14.288	[0.5625]
0.38	[0.015]	14.684	[0.5781]
0.76	[0.030]	15.081	[0.5937]
1.14	[0.045]	15.478	[0.6094]
ST-1235	Standard	15.875	[0.6250]
0.38	[0.015]	16.272	[0.6406]
0.76	[0.030]	16.669	[0.6562]
1.14	[0.045]	17.066	[0.6719]
ST-1236	Standard	17.463	[0.6875]
0.38	[0.015]	17.859	[0.7031]
0.76	[0.030]	18.256	[0.7187]
1.14	[0.045]	18.653	[0.7344]
ST-1237	Standard	19.050	[0.7500]
0.38	[0.015]	19.447	[0.7656]
0.76	[0.030]	19.844	[0.7812]
ST-1238		22.621	[0.8906]
		23.813	[0.9375]

The drill bushing that is used **must** be the same size as the reamer (or the drill) that is used.

⚠ CAUTION ⚠

Do not allow metal chips to enter the engine. Damage to the engine will result.



If the new dowel pins are more than 0.38 mm [0.015 in] larger than the old dowels, drill the hole to a size that is slightly smaller than the reamer. Then the reamer will **not** have to remove an excessive amount of material.

Ream the hole until the reamer touches the bottom of the hole in the block.

Remove the reamer.

Clean the hole and run the reamer through the hole again. The reamer **must** touch the bottom of the hole in the block.

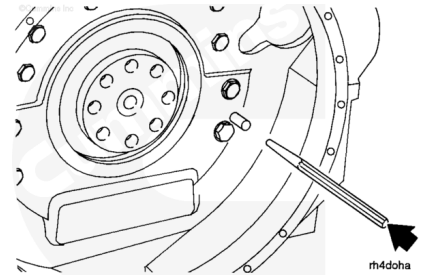
After reaming one hole, turn the plate and align it with the second dowel hole. Repeat the procedure in the second hole.

⚠ CAUTION ⚠

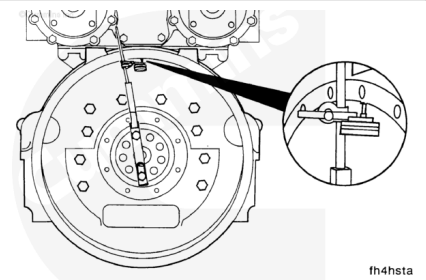
**Be sure the dowel hole does not contain any metal chips.
Engine damage will result.**

Remove the plate from the crankshaft.

Use a square nose drift. Drive each dowel in until it touches the bottom of the hole in the block.



After the dowels are installed, measure the bore and the face alignment again.

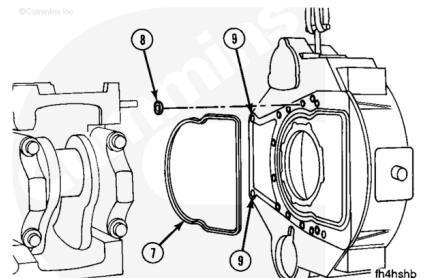


Install

Note : Be sure the flywheel housing dowels have been installed in the block.

Install the sealing ring (7) in the groove on the housing. If a wet type is used, install the seals (8) in the counterbores as shown. The holes (9) do **not** require seals.

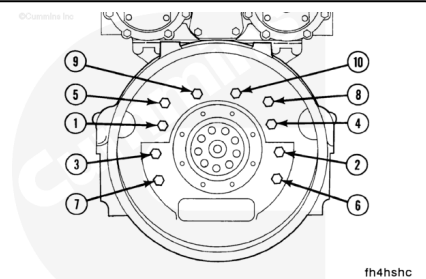
Use guide bolts to help during alignment. Install the housing and the capscrews.



Tighten the flywheel housing capscrews using the sequence shown.

Torque Value:

1. 100 n•m [75 ft-lb]
2. 205 n•m [150 ft-lb]

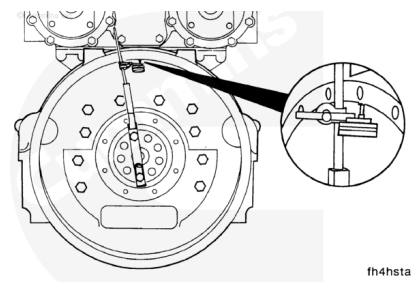


Measure the flywheel housing alignment.

Note : The bore and the face of the housing must be in alignment with the crankshaft.

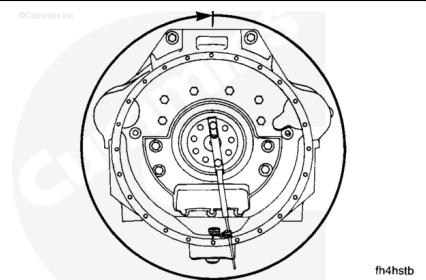
Note : The indicator arm must be rigid for an accurate reading. It must not sag.

Attach an indicator to the crankshaft as shown.



fh4hsta

Position the indicator at the 12 o'clock position. Adjust the dial until the needle points to ZERO. Rotate the crankshaft one complete revolution (360 degrees). Record the top indicator reading (TIR).

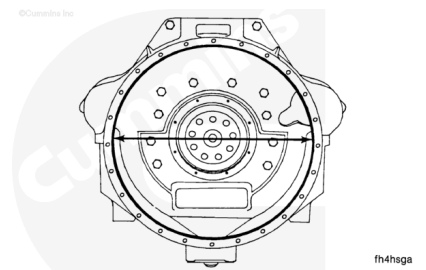


fh4hsib

The maximum allowable TIR depends on the diameter of the bore.

Bore Diameter/Maximum Top Indicator Reading (TIR)

Minimum to Maximum mm	Minimum to Maximum [in]	SAE	mm	[in]
787.4 to 810.5	[31.00 to 31.91]	00	0.30	[0.012]
647.7 to 648.0	[25.50 to 25.51]	0	0.25	[0.010]
584.2 to 584.4	[23.00 to 23.008]	1/2	0.25	[0.010]
511.2 to 511.3	[20.125 to 20.13]	1	0.20	[0.008]



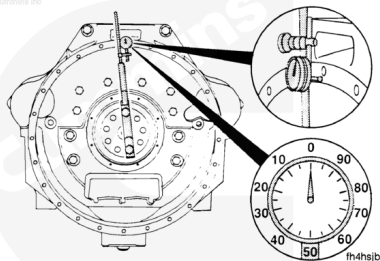
fh4hsge

If the alignment is **not** within specifications and the bore is round, the housing can be shifted.

If the alignment is **not** within specifications and the bore is **not** round, the housing **must** be replaced.

Note : The crankshaft end clearance must be pushed or pulled in the same direction each time a point is measured.

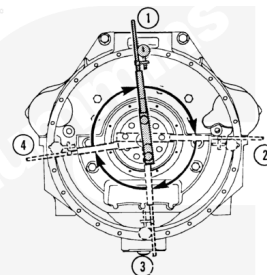
Attach an indicator as shown. Position the indicator at the 12 o'clock position. Adjust the dial until the needle points to ZERO.



Record the indicator reading at three different points; 3 o'clock, 6 o'clock, and 9 o'clock.

Turn backward to the original position. Be sure the needle still points to ZERO. Determine the total indicator runout (TIR).

		mm	[in]
Example:	3 o'clock	0.00	[0.00]
	6 o'clock	+0.08	[+0.003]
	9 o'clock	-0.05	[-0.002]
Equal TIR:		0.13	[0.005]

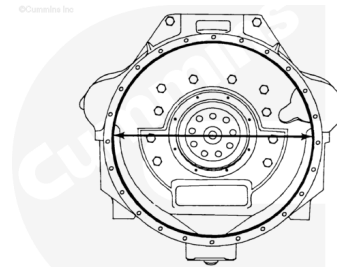


The maximum allowable TIR depends on the diameter of the bore.

Bore Diameter/Maximum TIR

Minimum to Maximum mm	Minimum to Maximum [in]	SAE	mm	[in]
787.4 to 810.5	[31.00 to 31.91]	00	0.30	[0.012]
647.7 to 648.0	[25.50 to 25.51]	0	0.25	[0.010]
584.2 to 584.4	[23.00 to 23.008]	1/2	0.25	[0.010]
511.2 to 511.3	[20.125 to 20.13]	1	0.20	[0.008]

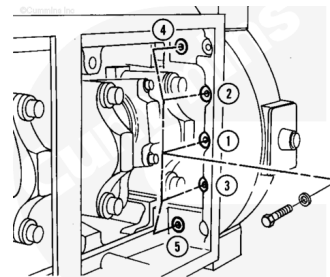
Note : If the alignment is not within specifications, remove the housing. Check for nicks, burrs, or foreign material between the block and the housing. Check the alignment again. If the alignment is not within specifications, the block or the housing is not machined correctly.



Tighten the five 9.5 mm [3/8-in] washers and capscrews in the sequence shown.

Torque Value:

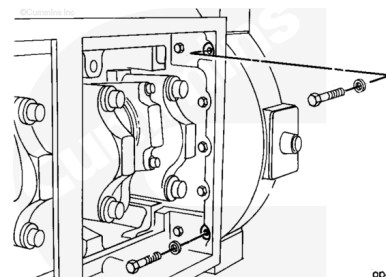
1. 25 n•m [20 ft-lb]
2. 40 n•m [30 ft-lb]
3. 45 n•m [35 ft-lb]



fh4csha

Install the two 11 mm [7/16-in] washers and capscrews as shown. Tighten the capscrews.

Torque Value: 65 n•m [48 ft-lb]

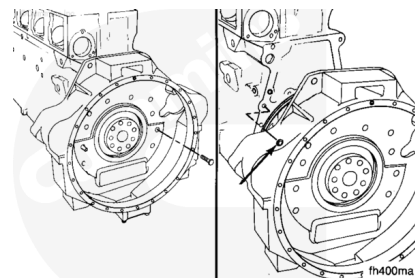


op4adh

Install the flywheel housing mounting capscrews.

Tighten the capscrews.

Torque Value: 205 n•m [150 ft-lb]

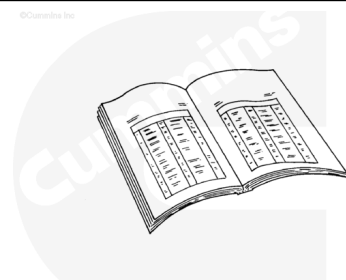


fh400ma

Maintenance Check

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, away ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the rear engine mounts to the flywheel housing. Refer to Procedure 016-010 in Section 16. (</qs3/pubsys2/xml/en/procedures/20/20-016-010.html>)
- Install the adapter cover plate or oil pan (whichever is in the front position). Refer to Procedure 007-026 in Section 7 (</qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html>) or Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-025-tr.html>)
- Install the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/20/20-013-020-tr.html>)
- Install the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-024-tr.html>)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/20/20-016-005-tr.html>)
- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Fill the engine with oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.